

Practical – 5

Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Theory :

1. Are correct input types and attributes used?

Yes, correct input types such as text, password, email, number, and date are used. The required attribute is also used to make important fields mandatory.

2. Are validation errors displayed near the relevant fields?

Partially, validation messages are displayed using alert() and some messages are displayed using JavaScript. For better usability, errors can be displayed directly near the relevant fields.

3. Is password strength checked?

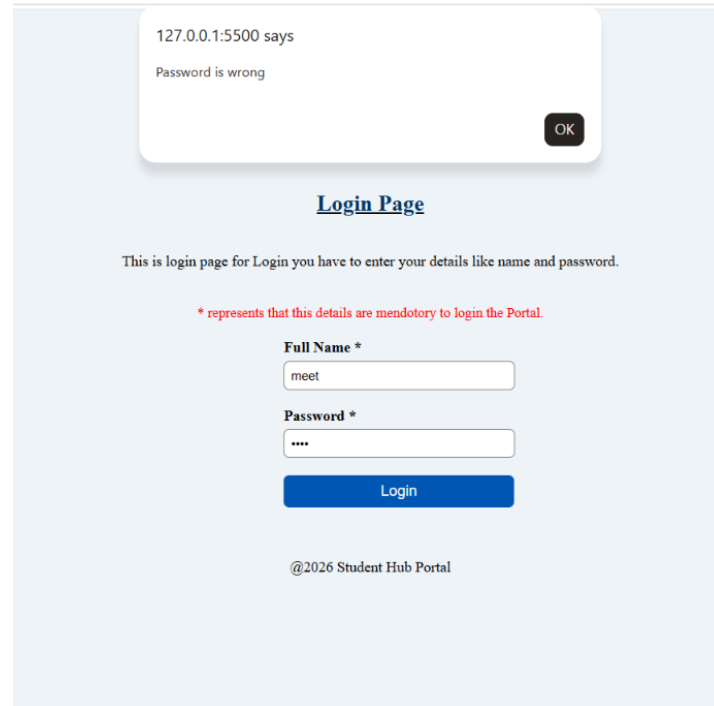
Yes, password strength is checked using Regex in JavaScript. The password must contain at least 8 characters, one letter, and one number.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

Yes, proper <label> tags with for attributes are used for the form fields. This helps users navigate the form using the keyboard and makes it more understandable for screen readers.

Implementation/ Output/Result validation of password and Email id using
Regex: (Pages with JS)

(1)Login page first try with wrong password:



127.0.0.1:5500 says
Password is wrong

Login Page

This is login page for Login you have to enter your details like name and password.

* represents that this details are mendotory to login the Portal.

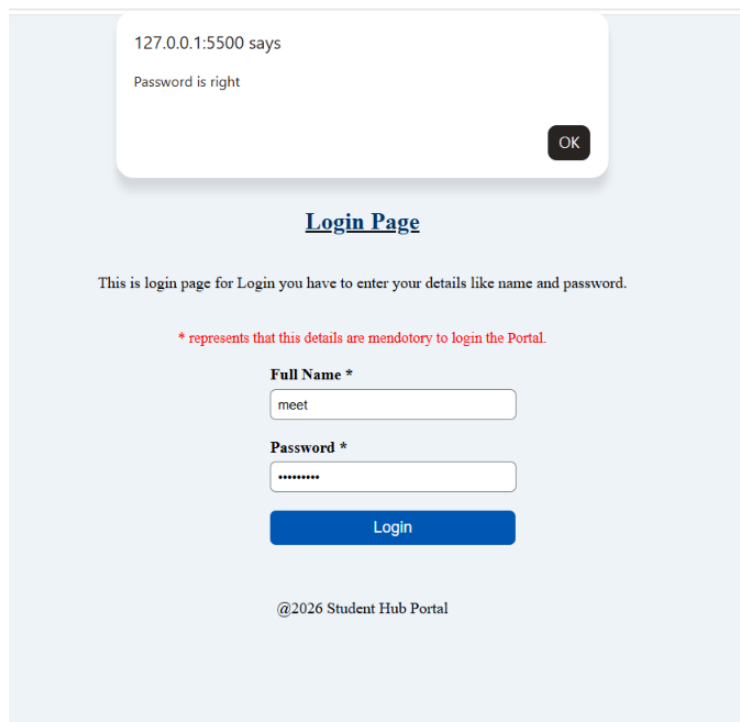
Full Name *

Password *

Login

@2026 Student Hub Portal

(2)Now try with right password:



127.0.0.1:5500 says
Password is right

Login Page

This is login page for Login you have to enter your details like name and password.

* represents that this details are mendotory to login the Portal.

Full Name *

Password *

Login

@2026 Student Hub Portal

(4)Now try with right password and right email.

127.0.0.1:5500 says
Password is right

OK

Password: *

Email ID: *
meet@1234gmail.com

Contact No: *
55441122

Address: *
hii

Pin Code: *
361007

Date of Birth: *
11/02/2000

Gender: *
Male

Department: *
Computer Engineering

Submit

127.0.0.1:5500 says
Email is right

OK

Password: *

Email ID: *
meet@1234gmail.com

Contact No: *
55441122

Address: *
hii

Pin Code: *
361007

Date of Birth: *
11/02/2000

Gender: *
Male

Department: *
Computer Engineering

Submit

Validation java script code using Regex in Login page:

```
77 // add regex for validation of password of user practical 5
78 let password = document.querySelector("#password");
79 let button = document.querySelector("#button_login");
80
81 let passwordPattern = /^(?=.*[A-Za-z])(?=.*[0-9]).{8,}$/;
82
83 button.addEventListener("click", function() {
84
85     if (passwordPattern.test(password.value)) {
86         alert("Password is right");
87     }
88     else {
89         alert("Password is wrong");
90     }
91 });
92 }
93
```

Validation java script code using Regex in Register page:

```
263 // for practical 5 add regex for validation of password of user and email id verification is here
264
265 let registerPasswordPattern = /^(?=.*[A-Za-z])(?=.*[0-9]).{8,}$/;
266 let registerEmailPattern = /^[A-Za-z0-9]+@[A-Za-z0-9]+\.[A-Za-z]{2,6}$/;
267
268 registerbutton.addEventListener("click", function() {
269
270     if (!registerPasswordPattern.test(password_register.value)) {
271         alert("Password is wrong");
272     }
273     else {
274         alert("Password is right");
275     }
276
277     if (!registerEmailPattern.test(email_register.value)) {
278         alert("Email is wrong");
279     }
280     else {
281         alert("Email is right");
282     }
283
284 });
285 }
```

GitHub Repository Link:

<https://github.com/MeetBhatti/WDF.git>

Learning Outcomes:

After completing this practical, I learned:

- To use **JavaScript DOM manipulation** for selecting and modifying HTML elements.
- To handle **user events** such as button clicks using `addEventListener()`.
- To perform **form validation** using JavaScript and **Regex** for password and email.
- To create a more **interactive and user-friendly** website.
- To improve the **usability, validation, and interactivity** of the Student Hub Portal.

Faculty Remarks: (Keep this blank for faculty)